Mess and Canteen Management System

Simple Project Guide for the whole team

1. What are we building?

A website for our college hostel mess and canteen. Everything is stored in one database, so the numbers always match: who ate, how much food was served, how much the caterer should be paid, and how much stock is left.

Why it matters: Today meals are counted by hand, so it is hard to check the caterer's bill, and nobody knows exactly how much to cook.

2. Who uses it?

Person	What they do in the system
Student	Applies for leave, orders from the canteen, gives feedback, raises complaints
Caterer / Manager	Plans the menu, manages stock, suppliers and staff, checks bills and costs
Mess / Canteen Staff	Marks who ate at the mess, bills canteen orders, enters stock
Warden / College Admin	Approves leave, checks meal counts, approves the caterer's payment

3. How it works in one day

- **Morning:** The system shows the caterer how many plates to cook (students enrolled minus students on approved leave).
- **Meal time:** Staff at the mess entrance scan or tick each student's ID. Each scan is one plate served.
- **Canteen:** Students order and pay. Stock goes down automatically.
- **Night:** The warden's office scans IDs for hostel attendance. This is only to check who is in the hostel, not to count meals.
- **End of month:** The system adds up all plates served, multiplies by the rate, and creates the caterer's bill. The warden checks it and approves payment.

4. How money works

Item	Simple explanation
Mess food	Free for students. No monthly bill to students.
Caterer's payment	The college pays the caterer **per plate served** (our assumption, still to confirm).
Guest meals	Not counted and not billed.
Canteen	Students pay for each order (cash, UPI or wallet).

Two different attendances: Hostel attendance (night, warden's office) and mess attendance (at meals). Only mess attendance is used for the bill. Hostel attendance is used to catch mistakes, for example a student marked as eating while on leave.

5. The 6 parts of the project

Part	What it does (in plain words)
1. Students and Leave	Stores student details and mess enrollment. Handles leave requests.
2. Attendance	Records who ate which meal. Flags suspicious entries.
3. Caterer Billing	Calculates plates x rate and creates the caterer's bill. Tracks approval and payment.
4. Menu and Canteen	Weekly mess menu, canteen items, student orders.
5. Inventory and Suppliers	Tracks stock, purchases from suppliers, daily usage and wastage.
6. Users, Feedback, Complaints	Logins and roles, meal ratings, complaint tracking until resolved.

6. DBMS words, explained simply

Our project must show these ideas. Each one is used in a real place in our system.

Word	Meaning	Example in our project
Table	A grid of data, like an Excel sheet	Student table, MealAttendance table
Primary key	A unique ID for each row	Every student has a unique student_id
Foreign key	A link from one table to another	An attendance row points to a student
Normalization	Splitting data so nothing is stored twice	Meal rates in their own table, not repeated in every attendance row
Constraint	A rule the database enforces	A student cannot be marked twice for the same meal
Join	Combining two tables in one query	Attendance + rates = bill amount
View	A saved query that acts like a table	v_daily_plates: plates served per day
Trigger	Runs automatically when data changes	Buying stock automatically increases stock count
Stored procedure	A saved set of steps you run with one command	generate_invoice creates the whole bill
Transaction	All steps succeed together, or none do	If an order fails halfway, nothing is saved
Index	Makes searches faster, like a book index	Index on meal date speeds up daily reports
Access control	Different users get different permissions	Students cannot see the caterer's bills

7. Who does what

Six people, six parts. Everyone builds the same amount: their own part of the database, at least one trigger, one stored procedure or transaction, one view, four queries, their own screens, one extra shared job, one report chapter and two PPT slides.

Person	Part	Extra shared job
Member 1	Students and Leave	Explain normalization (document how we split the data)

Person	Part	Extra shared job
Member 2	Attendance	Create realistic sample data for everyone to test with
Member 3	Caterer Billing	Make searches faster with indexes and show the improvement
Member 4	Menu and Canteen	PPT and demo script
Member 5	Inventory and Suppliers	Set up user permissions (who can see or change what)
Member 6 (Lead)	Users, Feedback, Complaints	Master ER diagram, joining everyone's work, final report

Member 1 - Students and Leave

You build	Tables for hostels, students, mess enrollment and leave requests
Trigger	Stops a student from applying for overlapping leave dates
Procedure	Approve a leave request and record who approved it
View	Students on leave today
Screens	Student profile, leave request form, leave approval page

Member 2 - Attendance

You build	Tables for meal timings, hostel attendance and mess attendance
Trigger	Blocks double entries, entries after the meal cutoff, and edits to already-billed days
Procedure	Mark a student's attendance for a meal
Views	Plates served per day; suspicious entries (ate while on leave, or not in hostel that day)
Screens	Attendance marking page, daily attendance report

Member 3 - Caterer Billing

You build	Tables for meal rates, caterer bills, bill lines and payments
Trigger	Forces the bill status to move in order: Generated, Verified, Approved, Paid
Procedure	Generate a bill for a date range (all steps in one transaction)
View	Bill summary
Screens	Generate bill, verify bill, approve and record payment

Member 4 - Menu and Canteen

You build	Tables for categories, menu items, weekly mess menu, orders and order items
Triggers	Save the price at the time of ordering; keep the order total correct
Procedure	Place an order (cancelled completely if an item is unavailable)
Views	Daily canteen sales; top selling items
Screens	Menus, cart and checkout, order history, menu management

Member 5 - Inventory and Suppliers

You build	Tables for stock items, suppliers, purchases, purchase items and daily usage
Triggers	Add stock on purchase; reduce stock on usage and reject negative stock; flag low stock
Procedure	Record a purchase (all steps in one transaction)
Views	Low stock items; money owed to suppliers
Screens	Stock dashboard, purchase entry, daily usage entry

Member 6 (Lead) - Users, Feedback, Complaints

You build	Tables for staff, user logins and roles, feedback, complaints and complaint history
Trigger	Records every complaint status change in a history table
Procedure	Resolve a complaint with a note
Views	Average rating per meal; complaints still open and how old they are
Screens	Login, role-based dashboards, feedback form, complaint tracker

8. Timeline (5 weeks)

Week	What happens
1	Confirm open questions with the warden. Everyone designs their part. Lead merges into one ER diagram.
2	Create tables and sample data. **Schema freeze:** no table changes after this week.
3	Write triggers, procedures, views and queries. Add indexes.
4	Build backend and screens. Join all parts together.
5	Testing, report, PPT, demo practice.

9. Still to confirm with the warden

- Is the caterer paid per plate, or a fixed monthly amount?
- Is the rate the same for all meals? Is there a minimum number of plates per day?
- How will mess attendance be taken, and by whom?
- Is hostel attendance on paper, Excel or software?
- Is there a canteen, and who runs it?
- What is the cutoff time for marking each meal?

10. Team rules

- One shared Git repository. Each person keeps their own SQL file.
- Table names start with capital letters (Student), column names use small letters with underscores (student_id).
- After the week 2 schema freeze, ask the Lead before changing any table.
- Weekly team meeting: each person reports progress on their part.